Use Case Flow - Event Management System

UC01 - User Role Registration and Dashboard Routing

Use Case ID	UC01
Use Case Name	User Role Registration and Dashboard Routing
Actor	Organizer / Attendee
Precondition	User has accessed the registration page and does not have an active session.
Main Flow	1. User navigates to the sign-up page. 2. Selects role: Organizer or Attendee. 3. Enters account credentials (name, email, secure password). 4. Submits the registration form. 5. System validates credentials and uniqueness of email. 6. System creates account and assigns role-based permissions. 7. System directs user to role-specific dashboard (Organizer event tools or Attendee browsing).
Alternative Flow	1. If required fields are omitted, system displays validation error. 2. If email already exists, system rejects submission and prompts login. 3. If an Attendee attempts to reach Organizer dashboard endpoints, system enforces RBAC and denies access.
Postcondition	User account is registered and authenticated with appropriate role dashboard.

UC02 - Create and Publish Event

Use Case ID	UC02
Use Case Name	Create and Publish Event
Actor	Organizer
Precondition	Organizer is authenticated with active Organizer session.
Main Flow	1. Organizer opens Event Management dashboard. 2. Clicks Create Event. 3. Enters title, date, start time, venue, ticket capacity, and ticket price. 4. Configures event category and description. 5. Saves draft or selects Publish. 6. System validates inputs and ensures capacity limit is positive and compliant with venue regulations. 7. System publishes event to public listing.
Alternative Flow	1. If mandatory details (date, price, capacity) are missing, system prevents publishing. 2. If event date is in the past, system rejects input with descriptive error. 3. If saving encounters database failure, system retains entered draft and alerts organizer.
Postcondition	Event is live, discoverable in search, and ready to accept ticket purchases.

UC03 - Browse and Filter Events

Use Case ID	UC03
Use Case Name	Browse and Filter Events
Actor	Attendee
Precondition	Platform contains published, active events.
Main Flow	1. Attendee visits event discovery page. 2. System loads and displays active upcoming events. 3. Attendee filters by date range, category, or searches by keyword. 4. System queries database and returns matching active events in under 2 seconds. 5. Attendee selects an event to view full details and ticket options.
Alternative Flow	1. If no events match query, system displays 'No active events found'. 2. Canceled, suspended, or past events are automatically omitted from search results. 3. If query fails due to connectivity, system displays graceful fallback message.
Postcondition	Attendee views detailed event page with available inventory count.

UC04 - Reserve Tickets and Hold Capacity

Use Case ID	UC04
Use Case Name	Reserve Tickets and Hold Capacity
Actor	Attendee
Precondition	Attendee is authenticated and selected event has available seats.
Main Flow	1. Attendee selects ticket quantity and clicks Checkout. 2. System initiates ACID-compliant transaction to check available capacity. 3. System places a temporary lock/hold on requested seats and decrements available count. 4. System initiates checkout countdown timer. 5. Attendee enters/confirms attendee details. 6. Attendee proceeds to payment stage.
Alternative Flow	1. If requested quantity exceeds remaining capacity, system blocks checkout and displays remaining count. 2. If concurrent checkouts compete for the last ticket, transactional lock permits one and informs the other. 3. If checkout timer expires, system releases held seats back to available capacity.
Postcondition	Ticket seats are held temporarily and user is transitioned to payment processing.

UC05 - Process Payment and Issue Digital Ticket

Use Case ID	UC05
Use Case Name	Process Payment and Issue Digital Ticket
Actor	Attendee / System / Payment Gateway
Precondition	Active ticket hold exists and Attendee is on payment screen.

Main Flow	1. Attendee enters payment details. 2. System tokenizes card data and communicates with external payment gateway via REST API. 3. Gateway approves charge and returns transaction token. 4. System converts temporary hold into permanent inventory reduction. 5. System generates unique digital ticket with UUID and QR code. 6. System emails confirmation receipt and ticket PDF to buyer. 7. System displays confirmation screen with ticket details.
Alternative Flow	1. If payment is declined, system releases held tickets back to available inventory. 2. System displays payment error message and allows user to retry or cancel. 3. Raw credit card data is never stored on platform databases (PCI-DSS compliant).
Postcondition	Payment is settled, inventory permanently updated, and digital ticket issued.

UC06 - Cancel Registration and Process Refund

Use Case ID	UC06
Use Case Name	Cancel Registration and Process Refund
Actor	Attendee
Precondition	Attendee holds a valid confirmed ticket for an upcoming event.
Main Flow	1. Attendee navigates to 'My Tickets' section. 2. Selects ticket and clicks Cancel Registration. 3. System verifies organizer's refund policy and event date. 4. Attendee confirms cancellation. 5. System voids unique Ticket ID and restores seat back to available capacity. 6. System triggers refund to original payment method via gateway. 7. System sends cancellation confirmation email.
Alternative Flow	1. If event has already commenced or policy disallows refund, cancellation is denied with clear explanation. 2. If refund gateway encounters network timeout, transaction is queued for automatic retry.
Postcondition	Ticket is voided, seat restored to inventory, and refund credited.

UC07 - Export Attendee Manifest (CSV)

Use Case ID	UC07
Use Case Name	Export Attendee Manifest (CSV)
Actor	Organizer
Precondition	Organizer is logged in and is the owner of the specified event.
Main Flow	1. Organizer opens event dashboard and selects specific event. 2. Clicks 'Download Attendee Manifest'. 3. System executes query joining user, ticket, and payment tables for confirmed attendees. 4. System excludes canceled and refunded tickets. 5. System generates structured CSV file containing names, emails, and ticket UUIDs. 6. Browser initiates download of clean CSV file.

Alternative Flow	1. If no tickets have been sold, system provides notification that manifest is currently empty. 2. If unauthorized user attempts to trigger manifest endpoint, system blocks request and logs security attempt.
Postcondition	Clean, verified attendee manifest CSV is downloaded for event check-in staff.

UC08 - Platform Admin Suspensions

Use Case ID	UC08
Use Case Name	Platform Admin Suspensions
Actor	Platform Admin
Precondition	Platform Admin is logged in with administrative privileges.
Main Flow	1. Admin accesses Platform Moderation dashboard. 2. Reviews flagged event or reported Organizer account. 3. Selects 'Suspend Event' or 'Suspend Organizer Account'. 4. Enters suspension rationale and confirms action. 5. System immediately delists event from public listings. 6. System terminates active checkouts and blocks further ticket sales. 7. System dispatches policy notification to Organizer.
Alternative Flow	1. If someone accesses direct URL of suspended event, system displays 'Event Suspended by Administrator'. 2. Admin can review and reinstate mistakenly suspended accounts after verification.
Postcondition	Fraudulent event/account is deactivated and ticket sales immediately halted.

Test Cases – Event Management System

Test Case ID	Req ID	Test Scenario	Preconditions	Test Steps	Expected Result	Priority	Status
TC01	FR-01	Register as Attendee with valid credentials	User is on registration page	1. Open Sign Up. 2. Select Attendee role. 3. Enter valid name, email, password. 4. Submit.	Account created and user routed to Attendee browsing dashboard.	High	Not Executed
TC02	FR-01	Register as Organizer with valid credentials	User is on registration page	1. Open Sign Up. 2. Select Organizer role. 3. Enter valid name, email, password. 4. Submit.	Account created and user routed to Organizer event creation dashboard.	High	Not Executed
TC03	FR-01	Prevent duplicate email registration	An account with email test@example.com exists	1. Fill registration with test@example.com. 2. Submit.	System rejects registration and displays 'Email already registered' error.	High	Not Executed
TC04	FR-01	Restrict Attendee from Organizer dashboard	User is logged in as Attendee	1. Manually navigate to /organizer/dashboard. 2. Attempt access.	Access denied (HTTP 403 / redirect to Attendee dashboard).	High	Not Executed
TC05	FR-02	Create and publish valid event	Organizer is logged in	1. Open Event Creator. 2. Enter title, date, time, venue, capacity (100), price ($20). 3. Click Publish.	Event is published and immediately visible on public listings.	High	Not Executed
TC06	FR-02	Reject event creation with missing capacity limit	Organizer is logged in	1. Fill event details leaving capacity empty. 2. Click Publish.	System blocks publication with 'Capacity limit is required' validation message.	High	Not Executed
TC07	FR-02	Prevent publishing event with past date	Organizer is logged in	1. Enter event details with date set to yesterday. 2. Click Publish.	System rejects event and displays date validation error.	High	Not Executed
TC08	FR-02	Update published event details	Event is published and Organizer is logged in	1. Open event editor. 2. Modify description and venue. 3. Save changes.	Event details updated successfully and reflected publicly.	Medium	Not Executed
TC09	FR-03	Filter events by category	Events exist across 'Tech' and 'Music' categories	1. Navigate to Events page. 2. Select 'Tech' category filter.	Only active events in the 'Tech' category are displayed.	Medium	Not Executed
TC10	FR-03	Filter events by date range	Active events scheduled across various dates exist	1. Select date range filter (e.g. Next 7 Days).	Only events scheduled within selected date range are returned.	Medium	Not Executed
TC11	FR-03	Exclude expired and canceled events from search	Expired and canceled events exist in database	1. Perform keyword search on main events page.	Expired and canceled events do not appear in default search results.	High	Not Executed
TC12	FR-04	Immediate ticket capacity decrement on checkout hold	Event has 50 tickets available	1. Attendee selects 2 tickets and clicks Checkout.	Available ticket count drops from 50 to 48 immediately.	High	Not Executed
TC13	FR-04	Block ticket request exceeding available capacity	Event has 1 ticket remaining	1. Attendee attempts to purchase 2 tickets.	System blocks checkout with 'Only 1 ticket remaining' alert.	High	Not Executed
TC14	FR-04	Release held seats on checkout timeout	Attendee placed 2 tickets on hold	1. Wait until checkout hold timer expires without paying.	Held tickets are released and event capacity returns to previous count.	High	Not Executed
TC15	FR-05	Validate contact details during checkout	Attendee is at checkout contact stage	1. Enter invalid email (missing @). 2. Click Continue.	System halts progression with 'Please enter a valid email address' alert.	Medium	Not Executed
TC16	FR-05	Accept valid attendee contact information	Attendee is at checkout contact stage	1. Enter valid full name, email, and phone number. 2. Click Continue.	Contact details verified and attendee proceeds to payment screen.	Medium	Not Executed

Test Case ID	Req ID	Test Scenario	Preconditions	Test Steps	Expected Result	Priority	Status
TC17	FR-06	Process successful payment and generate ticket	Tickets held, valid test payment token provided	1. Submit payment. 2. Gateway returns charge approval.	Unique Ticket UUID and QR code generated; confirmation email dispatched.	High	Not Executed
TC18	FR-06	Handle declined payment gracefully	Tickets held, card declined by gateway	1. Submit invalid card/declined charge.	Hold released back to inventory; error message displayed with retry option.	High	Not Executed
TC19	FR-06	Ensure unique UUID on generated tickets	Multiple concurrent successful ticket purchases	1. Verify generated ticket IDs in database.	Each issued ticket possesses a distinct, non-overlapping UUID.	High	Not Executed
TC20	FR-07	Cancel registration and void ticket ID	Attendee has active ticket for upcoming event	1. Open My Tickets. 2. Select ticket. 3. Click Cancel Registration. 4. Confirm.	Ticket ID is voided, capacity increases by 1, and confirmation is sent.	Medium	Not Executed
TC21	FR-07	Process refund to original payment method	Attendee cancels eligible ticket	1. Execute confirmed cancellation.	Refund is initiated through payment gateway back to original payment method.	Medium	Not Executed
TC22	FR-07	Prevent cancellation of past event ticket	Event finished yesterday	1. Attempt to click Cancel on finished event ticket.	System disables cancellation and displays 'Event has already concluded'.	Medium	Not Executed
TC23	FR-08	Export clean attendee manifest as CSV	Event has 15 confirmed attendees and 2 canceled	1. Organizer clicks 'Download Manifest'.	CSV downloads containing exactly the 15 confirmed attendees.	Medium	Not Executed
TC24	FR-08	Exclude canceled attendees from manifest	Event has canceled registrations	1. Inspect downloaded CSV manifest.	Canceled tickets and attendee records are completely excluded from manifest.	High	Not Executed
TC25	FR-08	Restrict manifest download to event owner	User B is not organizer of Event A	1. User B sends GET request to manifest endpoint for Event A.	Access denied (HTTP 403 Forbidden).	High	Not Executed
TC26	FR-09	Suspend fraudulent event by Platform Admin	Admin is logged into moderation console	1. Open reported event. 2. Click Suspend Event. 3. Confirm.	Event immediately disappears from public listings and sales are halted.	High	Not Executed
TC27	FR-09	Block direct URL checkout on suspended event	Event is in suspended status	1. Attendee visits direct event URL. 2. Attempt to checkout.	Checkout is blocked with 'This event has been suspended' notice.	High	Not Executed
TC28	FR-09	Suspend entire Organizer account	Organizer account violates terms of service	1. Admin clicks Suspend Organizer. 2. Confirm.	Organizer cannot log in and all their published events are de-listed.	High	Not Executed
TC29	NFR-01	Checkout response under simulated load	Simulate 500 concurrent checkout requests	1. Execute automated load test against checkout endpoint.	Sub-3-second response times maintained without server timeout.	High	Not Executed
TC30	NFR-01	Search page latency under standard traffic	100 active events in database	1. Perform event search and filter operations.	Search results returned in under 2 seconds.	High	Not Executed
TC31	NFR-02	Payment card tokenization verification	Attendee completes checkout with test credit card	1. Inspect database logs and storage records.	No plain-text credit card numbers stored; only gateway token and last 4 digits present.	High	Not Executed
TC32	NFR-02	PCI-DSS compliance log audit	Complete multiple test payment transactions	1. Audit application server logs for PAN/CVV.	Sensitive authentication data (CVV/PIN) is never logged or cached.	High	Not Executed
TC33	NFR-03	ACID concurrent booking transaction test	Event has exactly 1 remaining seat	1. Two attendees submit checkout simultaneously for the last seat.	ACID transaction guarantees exactly 1 purchase succeeds; 2nd gets graceful failure.	High	Not Executed

Test Case ID	Req ID	Test Scenario	Preconditions	Test Steps	Expected Result	Priority	Status
TC34	NFR-03	Rollback inventory hold on transaction failure	Database error simulated during ticket record creation	1. Complete payment authorization but trigger database exception.	Transaction rolls back safely; ticket count not corrupted.	High	Not Executed
TC35	NFR-04	Mobile viewport responsive checkout	Viewport set to mobile dimensions (375x667)	1. Complete full booking flow on mobile device emulator.	Layout adapts with touch-friendly buttons and no horizontal scrolling.	Medium	Not Executed
TC36	NFR-05	Payment gateway provider decoupling	Payment service abstraction layer in place	1. Switch payment gateway provider configuration.	Payment service logic switches without modifying core inventory database schema.	Low	Not Executed